Jayanth S

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My current work focuses on developing and evaluating AI/ML algorithms for physical layer research within Nokia's standards team supporting 3GPP activities. My research interests include Wireless Communication, Deep Learning, Reinforcement Learning, and Stochastic Optimization.

Education

Indian Institute of Technology, Dharwad — **2020 – 2022**
M.S(Research) in Electrical Engineering, CGPA = 9.17/10 — *Dharwad, Karnataka*
Thesis supervisor: Rajshekhar V Bhat

PES Institute of Technology, Bangalore South Campus — **2015 – 2019**
B.E. in Electronics and Communication Engineering, CGPA = 8.16/10 — *Bangalore, Karnataka*

Relevant Coursework

- Probability and Stochastic Processes
- Linear Algebra
- Convex Optimization
- Statistical Pattern Recognition
- Reinforcement Learning
- Wireless Communication

Experience

Nokia Standards — **Mar 2025 -**
Research Specialist — *Bangalore, Karnataka*
- Develop and run simulations to evaluate AI/ML algorithms for physical layer wireless communication research.
- Implement backend modules using language and retrieval models to build tools and pipelines for document analysis and data-driven research workflows.

TCS Research — **Feb 2023 - Mar 2025**
Research Engineer — *Bangalore, Karnataka*
- Implemented OFDM with RIS and OTFS modulation techniques using USRP B210 SDRs for data transmission experiments.
- Developed cascaded channel estimation algorithms for RIS(Reconfigurable Intelligent Surface) aided communication.
- Won the **Depth Estimation using mmWave AI/ML challenge - 2023** organised by NIST(National Institute of Standards and Technology, USA) and ITU(International Telecommunication Unit) **[Video Presentation]**.

Linköping University — **October 2022 - December 2022**
Research Assistant, Department of Computer and Information Science — *Linköping, Sweden*
- Explored **Open-loop Stationary Randomized Policy** framework for optimizing **semantics and age of information** with Assoc. Prof. **Nikolaos Pappas**.

TCS Research — **May 2022 - August 2022**
Research Intern — *Bangalore, Karnataka*
- Implemented **DDPG, TD3 and SAC** algorithms for Hybrid Beamforming in Single User - MIMO communication systems.

Projects and Paper Implementations

Policy Gradient Algorithms for Atari Games [code] — **2022**
- Applied **A2C, A3C, TRPO, and PPO**, based on the *stablebaselines3* and *ray rllib* implementations for Pong, Breakout and Space-Invaders atari games.

Implementation of the Paper - Model Free Training for End-to-End Communication Systems [code] — **2022**
- Implemented the model-based and model-free **auto-encoder** based end-to-end communication system for AWGN and Rayleigh Block Fading(RBF) channels as given in the paper: Fayçal Ait Aoudia and Jakob Hoydis, "Model-Free Training of End-to-End Communication Systems."

Resource Allocation For Overlay Device To Device(D2D) Communication — **2019**
- Implemented Largest Interference Aggregated First(LIFA) algorithm to utilize the resources available for effective device-to-device communication.

Technical Skills

Languages: Python, C/C++
Software Tools: Matlab, Latex
Packages/Frameworks used: TensorFlow, PyTorch, LangChain, Elasticsearch, Sionna, Ray RLlib
Applied AI/ML Techniques: Retrieval-Augmented Generation (RAG), LLM-based Backend Systems

Publications

1. **Jayanth S**, Vishnu Karthikeya Gorty, A Anil Kumar, Tapas Chakravarty, Arpan Pal, "RIBCE: RIS-BS virtual array based Channel Estimation for mmWave communication system," **IEEE ICC Workshop, 2024**.

2. **Jayanth S**, Nikolaos Pappas, Rajshekhar V Bhat, "Distortion Minimization with Age of Information and Cost Constraints," **WiOpt, 2023**.

3. **Jayanth S** and Rajshekhar V Bhat, "Age of Processed Information (AoPI) minimization with power constraint in fading multiple access channels," **IEEE ICC 2022** and the extended version of the paper is published in **IEEE Transactions on Wireless Communications(IEEE-TWC)** journal.

4. Gagan G B, **Jayanth S** and Rajshekhar V Bhat, "Age of Information Minimization with Power and Distortion Constraints in Multiple Access Channels," **WiOpt, 2021**.

Certified Courses

- Deep Learning Specialization- Coursera
- Reinforcement Learning Specialization - Coursera